HSC MATHEMATICS ADVANCED – YEAR 12 – TEACHER RESOURCE

Teaching Guide

The Limiting Sum of a Geometric Series

Topic: MA-M1 Modelling Financial Situations – M1.3 Geometric sequences and series

Includes: syllabus links, derivation, three key teacher demonstrations ($r = -\frac{1}{3}$, $r = \frac{1}{4}$ and why $r = 2$ fails), a two-lesson sequence, misconceptions, a practice set, HSC-style questions with marking guidelines and sample answers.

From the NESA Mathematics Advanced Reference Sheet (Financial Mathematics)

$$T_n = ar^{n-1} \qquad S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}, \quad r \neq 1 \qquad A = P(1+r)^n$$

$$\boxed{S = \frac{a}{1-r}, \qquad |r| < 1} \quad \text{(limiting sum)}$$

The reference sheet writes the limiting sum simply as S. Textbooks (and many students) write S_∞ – both are accepted in the HSC. ($A = P(1+r)^n$ is used in the perpetuity application, Q12.)

1. Overview for the teacher

Syllabus content (Mathematics Advanced Year 12, MA-M1, M1.3)

Students:

- derive and use the formula for the limiting sum of a geometric series with $|r| < 1$: $S = \dfrac{a}{1-r}$
- understand the limiting behaviour as $n \to \infty$ and its application to a geometric series as a limiting sum
- use the notation $\lim\limits_{n\to\infty} r^n = 0$ for $|r| < 1$.

Outcomes addressed

- **MA12-4** applies the concepts and techniques of arithmetic and geometric sequences and series in the solution of problems
- **MA12-2** models and solves problems and makes informed decisions about financial situations using mathematical reasoning and techniques (perpetuity application)
- **MA12-9** chooses and uses appropriate technology effectively (spreadsheet/calculator tables of partial sums)
- **MA12-10** constructs arguments to prove and justify results (explaining why a limiting sum does or does not exist)

Assumed prior knowledge

Geometric sequences and $T_n = ar^{n-1}$; the partial-sum formula S_n; index laws, including negative and fractional powers; absolute value notation ($|r| < 1 \iff -1 < r < 1$); solving linear and quadratic equations and inequalities; logarithms (for "how many terms" questions).

Suggested timing

Two 60-minute lessons. **Lesson 1:** concept of a limit, derivation, the three key demonstrations, guided practice. **Lesson 2:** applications (recurring decimals, bouncing ball, conditions on x, perpetuities) and HSC-style questions.

Learning intentions

Students will:

- understand what it means for the partial sums of a series to approach a limit;
- know *when* a geometric series has a limiting sum ($|r| < 1$) and *why*;
- use $S = \dfrac{a}{1-r}$ accurately, including for negative r;
- apply the limiting sum to practical and HSC-style problems.

Success criteria (student-facing)

"I can …"

- identify a and r and **check** $|r| < 1$ **before** using the formula;
- find a limiting sum, including when r is negative or a fraction;
- explain, with reasons, why a series such as $3 + 6 + 12 + \cdots$ has no limiting sum;
- work backwards: find a, r or a value of x from a given limiting sum.

2. Key ideas and the derivation

Vocabulary to teach explicitly

Partial sum S_n the sum of the first n terms: $S_n = T_1 + T_2 + \cdots + T_n$.

Limiting sum S **(or** S_∞**)** the value that S_n approaches as $n \to \infty$, if such a value exists. Also called the *sum to infinity*.

Converges / diverges a series *converges* if its partial sums approach a limit; otherwise it *diverges*.

Step 1 – What happens to r^n as n gets large?

Have students complete a table like this on their calculators (or in a spreadsheet) before any formula is shown.

n	$\left(\frac{1}{4}\right)^n$	$\left(-\frac{1}{3}\right)^n$	$(0.9)^n$	2^n
1	0.25	−0.3333	0.9	2
2	0.0625	0.1111	0.81	4
5	0.000 977	−0.004 12	0.590	32
10	9.5×10^{-7}	1.7×10^{-5}	0.349	1024
20	9.1×10^{-13}	2.9×10^{-10}	0.122	1 048 576
50	≈ 0	≈ 0	0.005	1.1×10^{15}

Conclusion: if $|r| < 1$ then $r^n \to 0$ as $n \to \infty$ (quickly for small $|r|$, slowly for $|r|$ near 1); if $|r| > 1$ then $|r^n|$ grows without bound. Introduce the notation:

$$\lim_{n \to \infty} r^n = 0 \quad \text{for } |r| < 1.$$

Step 2 – Derivation of the limiting sum formula

Start from the Reference Sheet partial-sum formula and let $n \to \infty$:

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r} = \frac{a\left(1-\lim\limits_{n\to\infty} r^n\right)}{1-r} = \frac{a(1-0)}{1-r} \qquad \text{(since } |r|<1\text{)}$$

$$\therefore \quad S = \frac{a}{1-r}, \qquad |r|<1.$$

The two things every student must say before using the formula

1. **Identify** a (first term) and $r = \dfrac{T_2}{T_1}$ (check with $\dfrac{T_3}{T_2}$).
2. **Check** $|r| < 1$, i.e. $-1 < r < 1$. Write it down: "*Since* $|r| = \frac{1}{3} < 1$, *a limiting sum exists.*" Markers look for this.

Teacher talk – the language of limits

Say "the partial sums get *as close as we like* to S" rather than "the sum *equals* S eventually". S_n never actually reaches S for a finite n; the *limit* of the partial sums is S. The useful error formula $S - S_n = \dfrac{ar^n}{1-r}$ (subtract the two formulas) shows the gap shrinks by a factor of r with every extra term – this is worth deriving with stronger classes.

3. Three key teacher demonstrations

The three examples below are deliberately chosen: a **negative** ratio, a **positive fractional** ratio, and a ratio **greater than 1**. Work each one as "I do" at the board, then have students reproduce the partial-sum table on their calculators.

Example 1 – A negative common ratio: $r = -\dfrac{1}{3}$

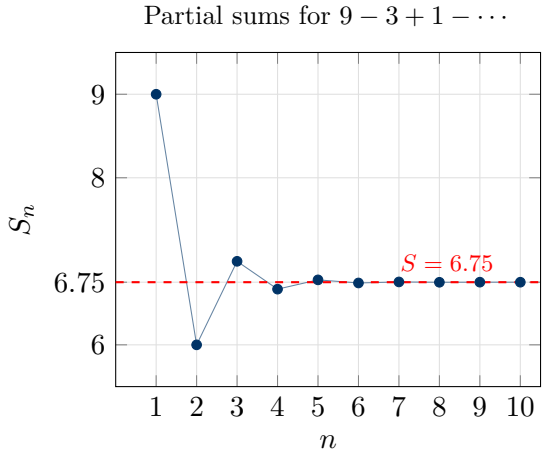
Find the limiting sum of the series $9 - 3 + 1 - \dfrac{1}{3} + \cdots$

Solution (NESA style)

$a = 9, \quad r = \dfrac{T_2}{T_1} = \dfrac{-3}{9} = -\dfrac{1}{3}$ (check: $\frac{T_3}{T_2} = \frac{1}{-3} = -\frac{1}{3}$ ✓)

Since $|r| = \dfrac{1}{3} < 1$, the limiting sum exists.

$$\begin{aligned} S &= \frac{a}{1-r} \\ &= \frac{9}{1-\left(-\frac{1}{3}\right)} \\ &= \frac{9}{\frac{4}{3}} = 9 \times \frac{3}{4} \\ &= \frac{27}{4} = 6\tfrac{3}{4} \end{aligned}$$



Answer: $S = \dfrac{27}{4} = 6.75$

Numerical check (partial sums):

n	1	2	3	4	5	6	7
S_n	9	6	7	6.667	6.778	6.741	6.753

Teaching points for Example 1

- The most common error is writing $1 - \frac{1}{3}$ in the denominator. Insist on the bracket: $1 - \left(-\frac{1}{3}\right) = 1 + \frac{1}{3} = \frac{4}{3}$.
- With a negative ratio the partial sums *oscillate* above and below S, closing in from both sides. The graph makes this visible.
- "Dividing by a fraction" is a Year 7–8 skill that resurfaces here – $\frac{9}{4/3} = 9 \times \frac{3}{4}$. Have students show this step.

Example 2 – A positive fractional ratio: $r = \frac{1}{4}$

Find the limiting sum of the series $16 + 4 + 1 + \frac{1}{4} + \cdots$

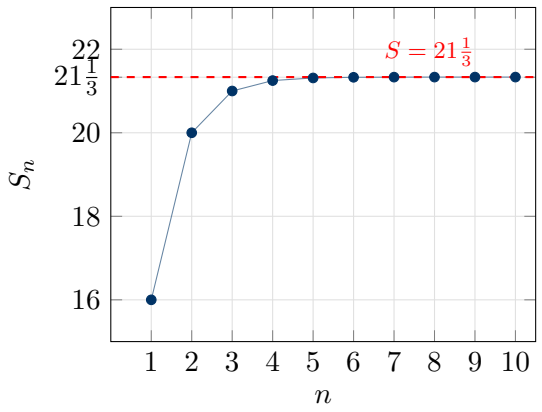
Solution (NESA style)

$a = 16, \quad r = \dfrac{T_2}{T_1} = \dfrac{4}{16} = \dfrac{1}{4}$ (check: $\frac{T_3}{T_2} = \frac{1}{4}$ ✓)

Since $|r| = \dfrac{1}{4} < 1$, the limiting sum exists.

$$\begin{aligned} S &= \frac{a}{1-r} \\ &= \frac{16}{1-\frac{1}{4}} \\ &= \frac{16}{\frac{3}{4}} = 16 \times \frac{4}{3} \\ &= \frac{64}{3} = 21\tfrac{1}{3} \end{aligned}$$

Partial sums for $16 + 4 + 1 + \cdots$



Answer: $S = \dfrac{64}{3} = 21\frac{1}{3}$

Numerical check (partial sums):

n	1	2	3	4	5	6	7
S_n	16	20	21	21.25	21.313	21.328	21.332

The gap $S - S_n = \dfrac{ar^n}{1-r} = \dfrac{64}{3}\left(\dfrac{1}{4}\right)^n$ is divided by 4 with each extra term.

Teaching points for Example 2

- With $0 < r < 1$ the partial sums increase and approach S *from below*: every $S_n < S$. Ask: "Can S_n ever equal $21\frac{1}{3}$?" (No – but it gets within any tolerance you name.)
- Link to Example 1: the *same* formula handles both signs of r; only the arithmetic in the denominator changes.
- Exact answers ($\frac{64}{3}$) are expected unless the question asks for a decimal.

Example 3 – Why $r = 2$ has *no* limiting sum

Does the series $3 + 6 + 12 + 24 + \cdots$ *have a limiting sum? Justify your answer.*

Solution (NESA style)

$a = 3, \quad r = \dfrac{6}{3} = 2.$

Since $|r| = 2 \geq 1$, the condition $|r| < 1$ is **not** satisfied, so the series does not have a limiting sum.

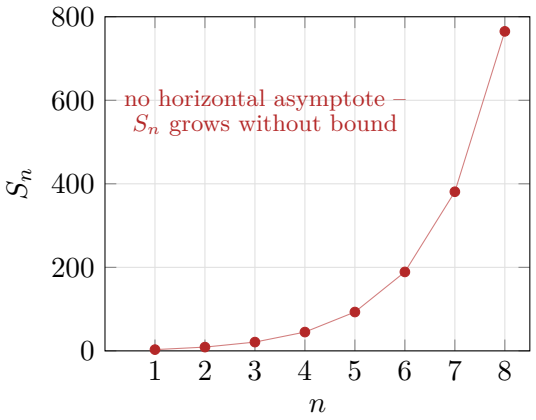
Why (what the partial sums actually do):

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{3(2^n - 1)}{2 - 1} = 3(2^n - 1)$$

n	1	2	3	4	5	6	10
S_n	3	9	21	45	93	189	3069

As $n \to \infty$, $2^n \to \infty$, so $S_n \to \infty$. The partial sums grow without bound and do not approach any fixed number: the series *diverges*.

Partial sums for $3 + 6 + 12 + \cdots$



Answer: No limiting sum, because $|r| = 2 \not< 1$ $(S_n = 3(2^n - 1) \to \infty)$.

What if a student uses the formula anyway?

Substituting blindly gives $\dfrac{a}{1 - r} = \dfrac{3}{1 - 2} = -3$. Ask the class: "Can adding positive numbers $3 + 6 + 12 + \cdots$ give -3?" The absurd answer shows that the formula is *only valid when* $|r| < 1$ – the derivation relied on $r^n \to 0$, which is false for $r = 2$ $(2^n \to \infty)$. This is the single most important lesson of the topic: **check** $|r| < 1$ **first.**

The boundary cases $r = 1$ and $r = -1$ (worth 2 minutes)

$r = 1$: $5 + 5 + 5 + \cdots$, $S_n = 5n \to \infty$ – no limiting sum (and the S_n formula itself is undefined, $r \neq 1$).

$r = -1$: $5 - 5 + 5 - 5 + \cdots$, S_n alternates $5, 0, 5, 0, \ldots$ – the partial sums never settle, so there is no limiting sum.

Hence the condition is the *strict* inequality $|r| < 1$, not $|r| \leq 1$.

4. Suggested lesson sequence

Lesson 1 – Concept, derivation and the three key examples (60 min)

Time	Phase	Activity
0–10	Hook	**Walking to the wall:** stand 8 m from the wall; each step covers half the remaining distance ($4 + 2 + 1 + \frac{1}{2} + \cdots$). "Do you ever arrive? What total distance do you approach?" Students vote, then calculate $S_1, \ldots, S_6$. (Alternative hook: is $0.\dot{9} = 1$?)
10–25	Explicit teaching	Table of r^n (Section 2, Step 1) on calculators; introduce $\lim_{n\to\infty} r^n = 0$ for $\lvert r\rvert < 1$; derive $S = \frac{a}{1-r}$ from S_n; state the condition.
25–45	Worked examples (I do / we do)	Example 1 ($r = -\frac{1}{3}$), Example 2 ($r = \frac{1}{4}$), Example 3 ($r = 2$ – no limiting sum). Students produce the partial-sum tables; teacher sketches the three graphs.
45–55	Guided practice	Practice Set A (Section 6), Q1–Q6, in pairs. Circulate and listen for "check $\lvert r\rvert < 1$".
55–60	Exit ticket	Three items (Section 6).

Lesson 2 – Applications and HSC-style problems (60 min)

Time	Phase	Activity
0–5	Recap	Quick-fire: "Limiting sum or not?" with $r = 0.7,\ -1.2,\ -\frac{1}{2},\ 1,\ \frac{5}{4}$.
5–25	Applications (I do / we do)	(i) Recurring decimal $0.\dot{4}\dot{5}$ as a GP; (ii) bouncing ball – total distance; (iii) series in x: for which x does a limiting sum exist?; (iv) perpetuity (link to MA-M1 financial maths).
25–50	HSC-style questions	Section 7, Q1–Q12. Students attempt individually, then peer-mark using the marking guidelines.
50–60	Reflection	Students write the "two things to say before using the formula" and one misconception they corrected. Collect for feedback.

Differentiation

Support: begin with $0 < r < 1$ only; provide the partial-sum table scaffold; use the "halving to the wall" model physically; supply a formula-check card ("a =? r =? $\lvert r\rvert < 1$?").

Extension: derive $S - S_n = \dfrac{ar^n}{1-r}$ and use logarithms to find the least n for a given accuracy (Q10); prove $0.\dot{9} = 1$; the trigonometric series in Q11; present value of a perpetuity (Q12) and its connection to the annuity formula.

5. Common misconceptions and how to address them

Misconception / error	**Teaching response**
Using $S = \frac{a}{1-r}$ when $\lvert r\rvert \geq 1$.	Example 3: the formula gives -3 for $3+6+12+\cdots$. Require the written statement "$\lvert r\rvert < 1$, so the limiting sum exists" in every solution.
Denominator sign error for negative r: writing $1-\frac{1}{3}$ instead of $1-\left(-\frac{1}{3}\right)$.	Always substitute with brackets. Check: a negative r makes the denominator *bigger than 1*, so S must be *smaller than* a (e.g. $6.75 < 9$).
"The series eventually reaches S" or "the last term is 0".	There is no last term; S is a *limit*. Use the gap formula $S - S_n = \frac{ar^n}{1-r} > 0$ to show S_n never equals S.
Misreading a in sigma notation, e.g. $\sum_{n=1}^{\infty} 6\left(\frac{2}{5}\right)^n$ taken as $a = 6$.	Write out the first two terms: $T_1 = \frac{12}{5}$, $T_2 = \frac{24}{25}$, so $a = \frac{12}{5}$, $r = \frac{2}{5}$, $S = 4$.
Solving $\lvert x-2\rvert < 1$ as $x < 3$ only.	$\lvert x-2\rvert < 1 \iff -1 < x-2 < 1 \iff 1 < x < 3$. Sketch the number line; test $x = 0$ (fails: $r = -2$).
Bouncing ball: forgetting that each rebound height is travelled *twice* (up and down), or doubling the initial drop as well.	Draw the path. Total = drop $+2\times$(sum of rebound heights).
Rounding too early (e.g. $r = 0.33$ for $\frac{1}{3}$).	Keep exact fractions until the final line; HSC answers are usually exact.
Confusing $r = \frac{T_2}{T_1}$ with $\frac{T_1}{T_2}$.	"Ratio = next term ÷ previous term"; check with a third term.

6. Practice Set A (fluency) and exit ticket

Practice Set A – find the limiting sum if it exists; if it does not, say why.

1. $8 + 4 + 2 + 1 + \cdots$
2. $18 - 6 + 2 - \frac{2}{3} + \cdots$
3. $1 + \frac{1}{3} + \frac{1}{9} + \cdots$
4. $100 + 10 + 1 + 0.1 + \cdots$
5. $2 + 3 + 4\frac{1}{2} + 6\frac{3}{4} + \cdots$
6. $0.3 + 0.03 + 0.003 + \cdots$
7. $5 - 5 + 5 - 5 + \cdots$
8. A GP has first term 12 and limiting sum 9. Find r.
9. A GP has common ratio $\frac{1}{4}$ and limiting sum 20. Find the first term.
10. Evaluate $\displaystyle\sum_{n=1}^{\infty} 6\left(\tfrac{2}{5}\right)^n$.

Answers

1. $r = \frac{1}{2}$; $S = \dfrac{8}{1 - \frac{1}{2}} = 16$
2. $r = -\frac{1}{3}$; $S = \dfrac{18}{\frac{4}{3}} = \dfrac{27}{2} = 13\frac{1}{2}$
3. $r = \frac{1}{3}$; $S = \dfrac{1}{\frac{2}{3}} = \dfrac{3}{2}$
4. $r = \frac{1}{10}$; $S = \dfrac{100}{0.9} = \dfrac{1000}{9} = 111\frac{1}{9}$
5. $r = \frac{3}{2}$, $|r| \geq 1$: no limiting sum
6. $r = \frac{1}{10}$; $S = \dfrac{0.3}{0.9} = \dfrac{1}{3}$
7. $r = -1$, $|r| \not< 1$: no limiting sum
8. $\dfrac{12}{1 - r} = 9 \Rightarrow 1 - r = \dfrac{4}{3} \Rightarrow r = -\dfrac{1}{3}$
9. $\dfrac{a}{1 - \frac{1}{4}} = 20 \Rightarrow a = 20 \times \dfrac{3}{4} = 15$
10. $a = \frac{12}{5}$, $r = \frac{2}{5}$; $S = \dfrac{12/5}{3/5} = 4$

Exit ticket (end of Lesson 1)

1. Find the limiting sum of $20 + 5 + 1\frac{1}{4} + \cdots$ *Answer:* $r = \frac{1}{4}$, $S = \dfrac{20}{3/4} = \dfrac{80}{3} = 26\frac{2}{3}$
2. Find the limiting sum of $6 - 2 + \frac{2}{3} - \cdots$ *Answer:* $r = -\frac{1}{3}$, $S = \dfrac{6}{4/3} = \dfrac{9}{2}$
3. Explain in one sentence why $1 + 3 + 9 + 27 + \cdots$ has no limiting sum. *Answer:* $r = 3$, $|r| \geq 1$, so $S_n \to \infty$

7. HSC-style questions (29 marks)

Instructions for students: Section I – choose the correct option. Section II – show all relevant working. The Reference Sheet is provided.

Section I – Multiple choice (1 mark each)

Question 1 Which of the following series has a limiting sum?
(A) $1 + 1.1 + 1.21 + \cdots$ (B) $1 - 1 + 1 - 1 + \cdots$ (C) $1 + 0.9 + 0.81 + \cdots$ (D) $1 + 2 + 3 + 4 + \cdots$

Question 2 The limiting sum of the series $12 + 3 + \dfrac{3}{4} + \cdots$ is
(A) 15 (B) 16 (C) 18 (D) 48

Question 3 A geometric series has first term 10 and limiting sum 8. The common ratio is
(A) $-\dfrac{1}{4}$ (B) $\dfrac{1}{4}$ (C) $-\dfrac{4}{5}$ (D) $\dfrac{4}{5}$

Section II – Short answer

Question 4 Find the limiting sum of the series $27 - 9 + 3 - 1 + \cdots$ **2**

Question 5 Explain why the series $5 + 10 + 20 + 40 + \cdots$ does not have a limiting sum. **2**

Question 6 The recurring decimal $0.\dot{4}\dot{5} = 0.454545\ldots$
(a) Write $0.\dot{4}\dot{5}$ as the sum of a geometric series, stating a and r. **1**
(b) Hence express $0.\dot{4}\dot{5}$ as a fraction in simplest form. **2**

Question 7 Consider the series $1 + (x-2) + (x-2)^2 + (x-2)^3 + \cdots$
(a) For what values of x does the series have a limiting sum? **1**
(b) Find the limiting sum in terms of x. **1**
(c) Hence find the value of x for which the limiting sum is 4. **1**

Question 8 A ball is dropped from a height of 10 metres. After each bounce it rises to $\frac{3}{5}$ of the height from which it fell. Find the total distance travelled by the ball before it comes to rest. **3**

Question 9 The second term of a geometric series is 6 and its limiting sum is 32. Find the possible values of the common ratio and the corresponding first terms. **3**

Question 10 Consider the series $16 + 4 + 1 + \frac{1}{4} + \cdots$ with limiting sum S.
(a) Show that $S - S_n = \frac{64}{3}\left(\frac{1}{4}\right)^n$. **2**
(b) Find the least value of n for which $S - S_n < 0.001$. **2**

Question 11 Show that, for $0 < x < \frac{\pi}{2}$, $\cos^2 x + \cos^2 x \sin^2 x + \cos^2 x \sin^4 x + \cdots = 1$. **3**

Question 12 A university wishes to establish a scholarship that pays \$6000 at the end of every year, forever. The fund earns interest at 5% per annum, compounded annually.
(a) Show that the amount invested now that provides the nth payment is $6000(1.05)^{-n}$. **1**
(b) Hence find the total amount that must be invested now to fund the scholarship indefinitely. **3**

8. Marking guidelines and sample answers

Section I

Q	Answer	Reasoning
1	C	$r = 0.9$ and $\lvert 0.9\rvert < 1$. (A) $r = 1.1$; (B) $r = -1$; (D) arithmetic, not geometric – all diverge.
2	B	$r = \frac{3}{12} = \frac{1}{4}$; $S = \dfrac{12}{1 - \frac{1}{4}} = \dfrac{12}{3/4} = 16$.
3	A	$\dfrac{10}{1-r} = 8 \Rightarrow 1 - r = \dfrac{10}{8} = \dfrac{5}{4} \Rightarrow r = -\dfrac{1}{4}$ (and $\lvert r\rvert < 1$ ✓).

Question 4 (2 marks)

Criteria	Marks
Provides correct answer	2
Identifies $a = 27$ and $r = -\frac{1}{3}$, or equivalent merit	1

Sample answer. $a = 27$, $r = \dfrac{-9}{27} = -\dfrac{1}{3}$. Since $|r| < 1$, the limiting sum exists.

$$S = \frac{a}{1-r} = \frac{27}{1-\left(-\frac{1}{3}\right)} = \frac{27}{\frac{4}{3}} = \frac{81}{4} = 20\tfrac{1}{4}$$

Question 5 (2 marks)

Criteria	Marks
Identifies $r = 2$ and explains that $\lvert r\rvert \geq 1$ (terms/partial sums increase without bound)	2
Identifies $r = 2$, or states the condition $\lvert r\rvert < 1$	1

Sample answer. $r = \dfrac{10}{5} = 2$. A limiting sum exists only when $|r| < 1$. Here $|r| = 2 > 1$, so $r^n = 2^n \to \infty$ and $S_n = \dfrac{5(2^n - 1)}{2 - 1} = 5(2^n - 1) \to \infty$. The partial sums grow without bound, so there is no limiting sum.

Question 6 (3 marks)

Criteria	Marks
(a) Writes $0.45 + 0.0045 + 0.000045 + \cdots$ with $a = 0.45$, $r = 0.01$	1
(b) Correct fraction $\frac{5}{11}$ with working	2
(b) Correctly substitutes into $S = \frac{a}{1-r}$, or equivalent merit	1

Sample answer.
(a) $0.\dot{4}\dot{5} = 0.45 + 0.0045 + 0.000045 + \cdots$, a geometric series with $a = 0.45$ and $r = 0.01$.
(b) Since $|r| < 1$: $S = \dfrac{0.45}{1 - 0.01} = \dfrac{0.45}{0.99} = \dfrac{45}{99} = \dfrac{5}{11}$.

Question 7 (3 marks)

Criteria	Marks
(a) Correct interval $1 < x < 3$	1
(b) Correct limiting sum $\frac{1}{3-x}$	1
(c) Correct value $x = \frac{11}{4}$	1

Sample answer.
(a) $a = 1$, $r = x - 2$. A limiting sum exists when $|x - 2| < 1$, i.e. $-1 < x - 2 < 1$, so $1 < x < 3$.
(b) $S = \dfrac{a}{1-r} = \dfrac{1}{1-(x-2)} = \dfrac{1}{3-x}$.
(c) $\dfrac{1}{3-x} = 4 \Rightarrow 3 - x = \dfrac{1}{4} \Rightarrow x = \dfrac{11}{4}$. Check: $1 < \frac{11}{4} < 3$ ✓, so the limiting sum exists.

Question 8 (3 marks)

Criteria	Marks
Correct total distance 40 m with working	3
Finds the limiting sum of the rebound heights (15 m), or sets up the total distance correctly	2
Identifies the rebound heights as a GP with $a = 6$, $r = \frac{3}{5}$, or equivalent merit	1

Sample answer. Rebound heights: 6, 3.6, 2.16, …, a GP with $a = 6$ and $r = \dfrac{3}{5}$ ($|r| < 1$).

$$\text{Sum of rebound heights} = \frac{6}{1 - \frac{3}{5}} = \frac{6}{\frac{2}{5}} = 15 \text{ m}$$

Each rebound height is travelled twice (up and down), and the initial 10 m drop once:

$$\text{Total distance} = 10 + 2 \times 15 = 40 \text{ m}$$

Question 9 (3 marks)

Criteria	Marks
Both values of r with corresponding first terms	3
Obtains the quadratic $16r^2 - 16r + 3 = 0$ (or equivalent) and one correct value of r	2
Writes $ar = 6$ and $\frac{a}{1-r} = 32$, or equivalent merit	1

Sample answer.

$$ar = 6 \qquad \cdots (1)$$

$$\frac{a}{1-r} = 32 \;\Rightarrow\; a = 32(1-r) \qquad \cdots (2)$$

Substitute (2) into (1): $32r(1-r) = 6$

$$32r - 32r^2 = 6$$

$$16r^2 - 16r + 3 = 0$$

$$(4r-1)(4r-3) = 0$$

$$r = \tfrac{1}{4} \text{ or } r = \tfrac{3}{4} \qquad \text{(both satisfy } |r| < 1)$$

When $r = \frac{1}{4}$: $a = 32\left(1 - \frac{1}{4}\right) = 24$. When $r = \frac{3}{4}$: $a = 32\left(1 - \frac{3}{4}\right) = 8$.

Question 10 (4 marks)

Criteria	Marks
(a) Correct derivation of $S - S_n = \frac{64}{3}\left(\frac{1}{4}\right)^n$	2
(a) Finds $S = \frac{64}{3}$ or writes S_n correctly	1
(b) Correct least value $n = 8$ with justification (logarithms or table)	2
(b) Sets up $\left(\frac{1}{4}\right)^n < \frac{3}{64000}$, or equivalent merit	1

Sample answer.

(a) $a = 16$, $r = \frac{1}{4}$, $|r| < 1$. $S = \dfrac{16}{1 - \frac{1}{4}} = \dfrac{64}{3}$ and $S_n = \dfrac{16\left(1 - \left(\frac{1}{4}\right)^n\right)}{1 - \frac{1}{4}} = \dfrac{64}{3}\left(1 - \left(\frac{1}{4}\right)^n\right)$.

$$S - S_n = \frac{64}{3} - \frac{64}{3}\left(1 - \left(\tfrac{1}{4}\right)^n\right) = \frac{64}{3}\left(\tfrac{1}{4}\right)^n \qquad \blacksquare$$

(b)

$$\frac{64}{3}\left(\tfrac{1}{4}\right)^n < 0.001$$

$$\left(\tfrac{1}{4}\right)^n < \frac{0.003}{64} = 0.000\,046\,875$$

$$n\log_{10}\left(\tfrac{1}{4}\right) < \log_{10}(0.000\,046\,875)$$

$$n > \frac{\log_{10}(0.000\,046\,875)}{\log_{10}\left(\frac{1}{4}\right)} = 7.19\ldots \qquad \text{(inequality reverses: } \log_{10}\tfrac{1}{4} < 0\text{)}$$

$\therefore$ the least value is $n = 8$. Check: $n = 7$ gives $S - S_7 = 0.0013 > 0.001$; $n = 8$ gives $0.00033 < 0.001$ ✓.

Question 11 (3 marks)

Criteria	Marks
Complete proof, including justification that $\lvert r\rvert < 1$	3
Identifies $a = \cos^2 x$, $r = \sin^2 x$ and applies $S = \frac{a}{1-r}$	2
Identifies $a = \cos^2 x$ and $r = \sin^2 x$	1

Sample answer. The series is geometric with $a = \cos^2 x$ and $r = \dfrac{\cos^2 x \sin^2 x}{\cos^2 x} = \sin^2 x$.

For $0 < x < \dfrac{\pi}{2}$, $0 < \sin x < 1$, so $0 < \sin^2 x < 1$; hence $|r| < 1$ and the limiting sum exists.

$$S = \frac{a}{1-r} = \frac{\cos^2 x}{1 - \sin^2 x} = \frac{\cos^2 x}{\cos^2 x} = 1 \qquad \text{(using } \sin^2 x + \cos^2 x = 1\text{)} \qquad \blacksquare$$

Question 12 (4 marks)

Criteria	Marks
(a) Uses $A = P(1.05)^n$ with $A = 6000$ to obtain $P = 6000(1.05)^{-n}$	1
(b) Correct total \$120 000 with working	3
(b) Identifies the GP with $a = \frac{6000}{1.05}$, $r = \frac{1}{1.05}$ and applies the limiting sum formula	2
(b) Writes the total as a series of present values, or equivalent merit	1

Sample answer.

(a) An amount P invested now grows to $A = P(1.05)^n$ after n years. For this to provide the nth payment, $P(1.05)^n = 6000$, so $P = 6000(1.05)^{-n}$. $\blacksquare$

(b) Total to invest now

$$= 6000(1.05)^{-1} + 6000(1.05)^{-2} + 6000(1.05)^{-3} + \cdots$$

a geometric series with $a = \dfrac{6000}{1.05}$, $r = \dfrac{1}{1.05} \approx 0.952$, $|r| < 1$

$$= \frac{a}{1-r} = \frac{\frac{6000}{1.05}}{1 - \frac{1}{1.05}} = \frac{6000}{1.05 - 1} \qquad \text{(multiplying numerator and denominator by 1.05)}$$

$$= \frac{6000}{0.05} = \$120\,000$$

Why this is a good HSC question

It links the limiting sum to MA-M1 financial mathematics: the present value of a *perpetuity* is payment ÷ interest rate ($6000 \div 0.05$). Students can verify the answer: \$120 000 earns 5% = \$6000 each year, which is exactly the payment, so the capital is never depleted.

One-page summary for students

The Limiting Sum of a Geometric Series

Formula (NESA Reference Sheet)

$$S = \frac{a}{1-r}, \qquad |r| < 1 \quad \text{(i.e. } -1 < r < 1\text{)}$$

where a is the first term and $r = \dfrac{T_2}{T_1}$ is the common ratio. It comes from $S_n = \dfrac{a(1-r^n)}{1-r}$ and $\lim\limits_{n\to\infty} r^n = 0$ for $|r| < 1$.

Series	a **and** r	$\lvert r\rvert < 1$**?**	**Limiting sum**
$9 - 3 + 1 - \frac{1}{3} + \cdots$	$a = 9,\ r = -\frac{1}{3}$	Yes ($\frac{1}{3} < 1$)	$\dfrac{9}{1-(-\frac{1}{3})} = \dfrac{27}{4}$
$16 + 4 + 1 + \frac{1}{4} + \cdots$	$a = 16,\ r = \frac{1}{4}$	Yes ($\frac{1}{4} < 1$)	$\dfrac{16}{1-\frac{1}{4}} = \dfrac{64}{3}$
$3 + 6 + 12 + 24 + \cdots$	$a = 3,\ r = 2$	**No** ($2 \geq 1$)	None – $S_n = 3(2^n - 1) \to \infty$
$5 - 5 + 5 - 5 + \cdots$	$a = 5,\ r = -1$	**No** ($1 \not< 1$)	None – S_n alternates $5, 0, 5, 0, \ldots$

Three steps every time

1. Identify a and r (check r with a second pair of terms).
2. Check and *state* that $|r| < 1$. If not, there is no limiting sum – explain why (r^n does not approach 0).
3. Substitute into $S = \dfrac{a}{1-r}$, using brackets when r is negative; give an exact answer.

Useful extras

- Gap after n terms: $S - S_n = \dfrac{ar^n}{1-r}$ (use logarithms to find how many terms give a required accuracy).
- Working backwards: $S = \dfrac{a}{1-r} \Rightarrow a = S(1-r)$ and $r = 1 - \dfrac{a}{S}$.
- Recurring decimals: $0.\dot{4}\dot{5} = \dfrac{0.45}{1-0.01} = \dfrac{5}{11}$.
- Bouncing ball: total distance $=$ drop $+ 2 \times \dfrac{\text{first rebound height}}{1-r}$.
- Perpetuity: amount needed now $= \dfrac{\text{payment}}{\text{interest rate}}$ (a limiting sum of present values).

Answers to the HSC-style questions (Section 7)	
1–3	C, B, A
4	$S = \frac{81}{4} = 20\frac{1}{4}$
5	$r = 2$, $\lvert r\rvert \geq 1$, so $S_n = 5(2^n - 1) \to \infty$: no limiting sum
6	(a) $a = 0.45$, $r = 0.01$ (b) $\frac{5}{11}$
7	(a) $1 < x < 3$ (b) $\frac{1}{3-x}$ (c) $x = \frac{11}{4}$
8	40 m
9	$r = \frac{1}{4},\ a = 24$ or $r = \frac{3}{4},\ a = 8$
10	(a) shown (b) $n = 8$
11	$a = \cos^2 x$, $r = \sin^2 x$ $(0 < r < 1)$; $S = \frac{\cos^2 x}{1-\sin^2 x} = 1$
12	(a) shown (b) \$120 000